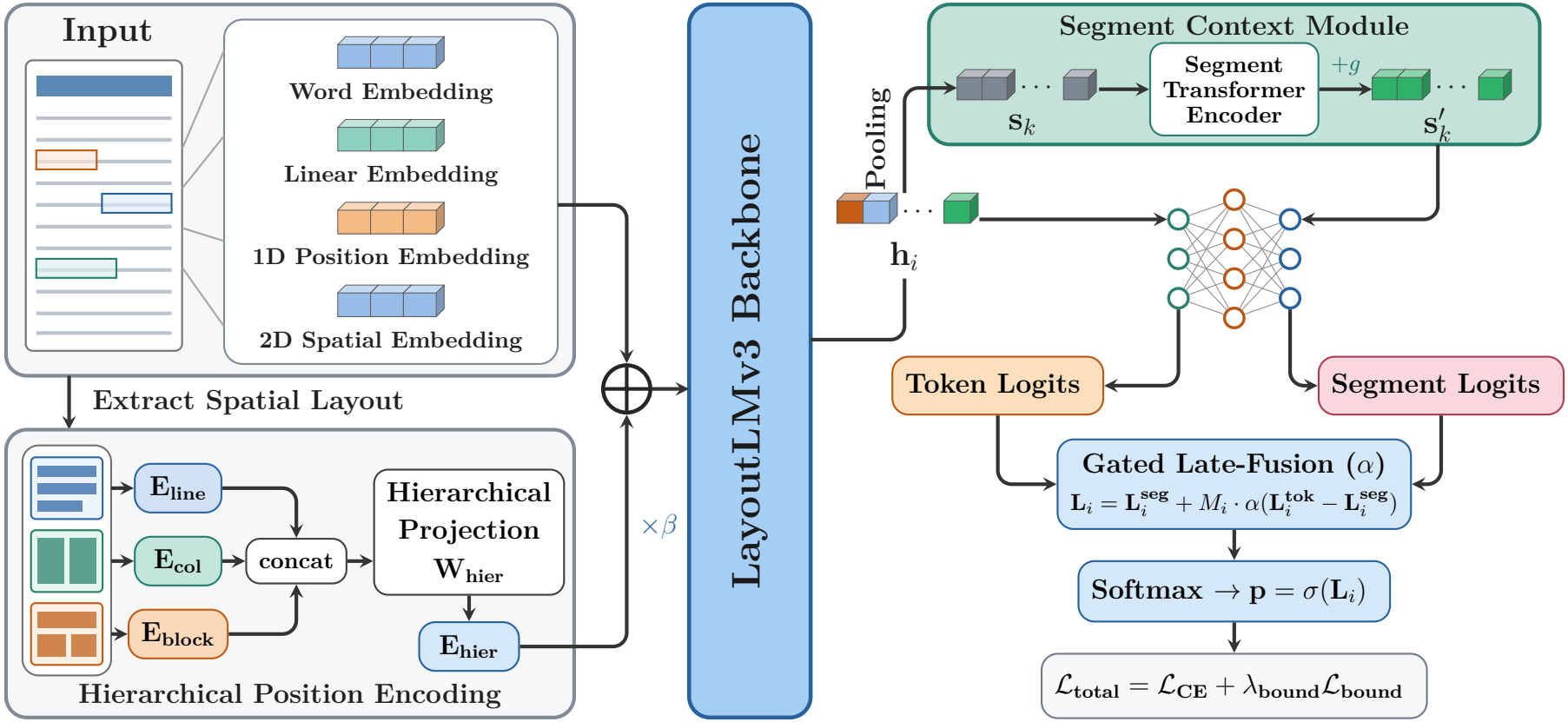
Input
Word Embedding
Linear Embedding
1D Position Embedding
2D Spatial Embedding
Extract Spatial Layout
$\mathbf{E}_{\text{line}}$
$\mathbf{E}_{\text{col}}$
$\mathbf{E}_{\text{block}}$
concat
Hierarchical Projection $\mathbf{W}_{\text{hier}}$
$\mathbf{E}_{\text{hier}}$
Hierarchical Position Encoding
$\times\beta$
LayoutLMv3 Backbone
Pooling
$\mathbf{h}_i$
Segment Context Module
$\mathbf{s}_k$
Segment Transformer Encoder
$+g$
$\mathbf{s}'_k$
Token Logits
Segment Logits
Gated Late-Fusion (α)
$\mathbf{L}_i = \mathbf{L}_i^{\text{seg}} + M_i \cdot \alpha(\mathbf{L}_i^{\text{tok}} - \mathbf{L}_i^{\text{seg}})$
Softmax $\rightarrow \mathbf{p} = \sigma(\mathbf{L}_i)$
$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda_{\text{bound}}\mathcal{L}_{\text{bound}}$